המרכז
לחינוך סייבר
CYBER EDUCATION CENTER

Assembly for Reverse Engineering

Introduction

Barak Gonen

Assembly Language

המרכז לחינוך סייבר
CYBER EDUCATION CENTER

- Machine language: 0100 1110 1111 0001 …
- Assembly instructions are slightly more human readable: mov, add, inc, mul …
- Writing assembly code requires deep knowledge of CPU (registers, flags, memory addresses)
- Assembly preceded High Level Languages
  - C, Java, Python
  - “Hide” the CPU internals from the programmer

Why Study Assembly?

המרכז לחינוך סייבר
CYBER EDUCATION CENTER

- … Want the job? :-)
- Plus it’s sooooo fun!

Malware Research Team Leader

Check Point Software Technologies Ltd · Tel Aviv, Israel (Hybrid)

Desired Background:

- Vast experience in hands-on security research related role
- Understanding the malware landscape - threat actors, malware groups, types of malware
- Experience in reverse-engineering software through static and dynamic analysis - IDA Pro, x64dbg, Ghidra, Cutter
- Knowledge of assembly languages - x86 or ARM
- Deep understanding of operating system concepts - preferred Windows.
- Ability to develop malware tracking and analysis tools - preferably in Python.

Course Objective

המרכז לחינוך סייבר
CYBER EDUCATION CENTER

- Feel comfortable with reading assembly code
- Required for Reverse Engineering

```
call    read_hex
add     eax, eax
mov     ebx, eax
shl     eax, ?
add     ebx, eax
shl     eax, 4
add     ebx, eax
call    print_eax
```